

Isometries Summary Sheet

Geometric data needed:

- Reflection: needs an axis/line in E^2 or a plane in E^3 .
- Rotation: needs a center/axis and an angle.
- Translation: needs a translation vector.
- Glide reflection: needs a reflection axis and a translation vector parallel to that axis.

1. Reflection

Reflection in the x-axis:

$$(x,y) \rightarrow (x,-y)$$

Matrix form:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Reflection in a plane with unit normal n :

$$\text{Ref}_{\perp n}(P) = (I - 2 n \otimes n) P$$

2. Glide Reflection

Reflect first, then translate.

Example:

$$(x,y) \rightarrow (x,-y) \rightarrow (x+2,-y)$$

Final formula:

$$(x,y) \rightarrow (x+2,-y)$$

Matrix + vector form:

$$X' = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} X + \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

3. Rotation

Rotation around the origin by angle θ :

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Needs:

- center
- angle
- in E^3 also an axis

4. Translation

Translation by vector $v=(a,b)$:

$$(x,y) \rightarrow (x+a,y+b)$$

Vector form:

$$X' = X + v$$

5. Direct vs Indirect Isometries

Direct isometries preserve orientation:

- rotations
- translations

$$\det(A)=+1$$

Indirect isometries reverse orientation:

- reflections
- glide reflections

$$\det(A)=-1$$

Orientation means preserving clockwise/counterclockwise order (handedness).